From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

ANSWERS, HINTS & SOLUTIONS
FULL TEST –I
(Main)

S. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	A	C	C
2.	B	A	B
3.	D	B	B
4.	B	D	C
5.	C	C	C
6.	C	B	A
7.	A	D	A
8.	C	B	C
9.	B	C	B
10.	C	D	A
11.	B	D	C
12.	D	A	D
13.	C	C	C
14.	B	D	B
15.	B	C	D
16.	D	C	B
17.	C	D	A
18.	D	B	B
19.	D	D	B
20.	D	C	D
21.	D	A	C
22.	B	A	A
23.	A	D	C
24.	A	D	B
25.	A	C	B
26.	A	C	B
27.	D	A	B
28.	B	A	B
29.	D	B	B
30.	B	B	D

Physics

PART – I

SECTION – A

1. For full square about an axis passing through 'O' $= \frac{M\ell^2}{6}$

   by symmetry for remaining portion it must be $\frac{3}{4}\left[\frac{M\ell^2}{6}\right] = \frac{M\ell^2}{8}$

2. apply work-energy theorem

   $$mgz - \left[\frac{1}{2}K\left(\frac{mg}{K}+z\right)^2 - \frac{1}{2}K\left(\frac{mg}{K}\right)^2\right] + Fz = 0$$

   $\Rightarrow z = 2F/K.$

3. $a = 2 + |t - 2|$
   for $t \le 2$
   $a = 2 - t + 2$
   $a = 4 - t$
   $dv = (4 - t)dt$
   $v = 4t - t^2/2$
   at t = 2, v = 6 m/s.
   for $t > 2$
   $a = 2 + t - 2 = t$
   $\int_6^v dv = \int_2^t t\,dv$
   $v - 6 = \left[t^2/2\right]_2^t$
   $v = \frac{t^2}{2} + 4$
   at t = 4, v = 12 m/s.

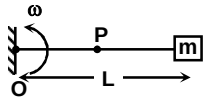
10. $(\vec{a}\times\vec{b}) \perp [\vec{a}\times(\vec{a}\times\vec{b})]$. Dot product is zero.

11. Say speed of boat is v w.r.t. water and speed of river is C. Then, distance travelled in ground frame

    $= (c + v) \times \frac{1}{2}$ hour $+ (v - c) \times \frac{1}{2}$ hour $= v \times 1$ hour

    = distance travelled by boat w.r.t. river.

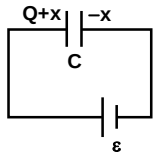
12. In the shown frame the particle appears to be at rest.
    ∴ Net force on it must be zero. Therefore pseudo force must be equal and opposite to the tension.'



13. Successive frequencies will differ by an amount v/2L

14. Linear magnification = v^2/u^2

15. Electric field between the capacitor plates = $\frac{\sigma_1}{2\varepsilon_0}+\frac{(-\sigma_2)}{2\varepsilon_0}$



$E = \frac{Q+x}{2A\varepsilon_0}+\frac{x}{2A\varepsilon_0}=\frac{1}{2A\varepsilon_0}[Q+2x]$

$\Rightarrow$ Potential different $E_d = \frac{d}{2A\varepsilon_0}[Q+2x]=\varepsilon$

$\Rightarrow \varepsilon = \frac{Q+2x}{2C}$

$\Rightarrow -x = \frac{Q}{2}-C\varepsilon$

16. conservation of momentum

$M\sqrt{2gh}-m\sqrt{2gh}=MV_1+mV_2$ (1)

$-1=\frac{V_1-V_2}{2\sqrt{2gh}}$ (2)

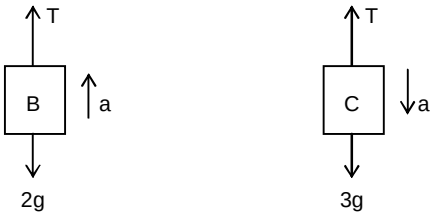
$V_2=3\sqrt{2gh}$

$h'=\frac{V_2^2}{2g}=9h$

17. $\frac{1}{C}=\frac{1}{C_1}+\frac{1}{C_2}$

$=\frac{x}{\varepsilon_0 A}+\frac{a-b-x}{\varepsilon_0 A}\Rightarrow C=\frac{\varepsilon_0 A}{a-b}$

18. FBD of 'B' and 'C'

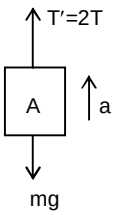


$\Rightarrow$ T – 2g = 2a . . . (i)

and 3g – T = 3a . . . (ii)

$T=\frac{12g}{5}$

for A



$\Rightarrow$ T = mg

$2\times\frac{12g}{5}=mg \Rightarrow m = 4.8$ kg.

19. $W = \int_0^{\pi/2} f.Rd\theta = \frac{\mu mgR\pi}{2}$ = 1 joule.

20. Coulombic force between them remains same.

21. Let us assume cylinder is no moving then

$T + f_s = mg \sin\theta$

$T.R - f_s R = 0$

$\Rightarrow f_s = \frac{mg\sqrt{3}}{4}$.

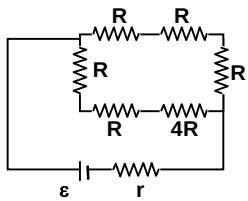
but $(f_s)_{max} = \mu N = \mu mg \cos\theta$

$= 0.4 . mg \times \frac{1}{2} = \frac{mg}{5}$

$\therefore$ $(f_s) < (f_s)_{max}$, our assumption is wrong. So, friction existing must be kinetic

$f_k = \mu mg \cos\theta = 0.4 \times mg \times \frac{1}{2} = \frac{mg}{5}$

22. Circuit is forming a wheatstone bridge $R_{eq} = 2R$
For maximum power transfer 2R = r.



23. $\phi = \frac{\sqrt{3}}{4}\ell^2 B$

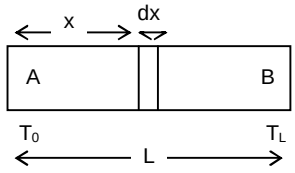
$\varepsilon = \left|\frac{d\phi}{dt}\right| = \frac{\sqrt{3}}{4}\ell^2 \frac{dB}{dt}$

$i = \frac{\varepsilon}{R} = \frac{\sqrt{3}\ell^2}{4R}$

24. $C = \sqrt{\frac{\gamma RT}{M_0}} = \sqrt{\frac{5RT}{3M_0}}$

$dx = C.dt = \sqrt{\frac{5R}{3M_0}\left[T_0 + \left(\frac{T_L - T_0}{L}\right)x\right]}dt$

$t = \frac{2L}{(\sqrt{T_L} + \sqrt{T_0})}\sqrt{\frac{3M}{5R}}$



25. $\frac{E}{2} = E - ir$

$\Rightarrow i = \frac{E}{2r}$ . . . (i)

$2E = i(3 + r)$ . . . (ii)

$\Rightarrow r = 1\ \Omega$

26. $dQ = dU + dW$

$C = C_v + \frac{PdV}{ndT}$ …(i)

Differentiating $TV^2 = \text{constant}$

$\frac{dV}{dT} = -\frac{V}{2T}$ …(ii)

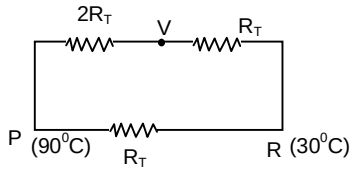
$PV = nRT$ …(iii)

solving eq. (i), (ii) and (iii)

$C = \frac{R}{2}$

27. Equivalent circuit is

$\frac{90 - V}{2R_T} = \frac{20}{R_T} \Rightarrow V = 50^0C$



Chemistry

PART – II

SECTION – A

2. 2B atoms (ions are missing from every unit cell).

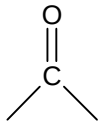
3. Only KI produces I^- ions in aq. solution.

6. Time period $\propto \frac{n^3}{z^3}$

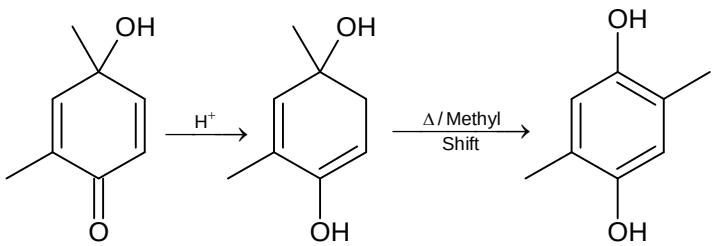
$\frac{T_1}{T_2} = \frac{n_1^3 \times z_2^3}{n_2^3 \times z_1^3}$; $z_1 = z_2$ for same element.

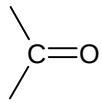
Hence, $\frac{T_1}{T_2} = \frac{1^3}{3^3} = \frac{1}{27}$

7. From K_{eq} value; % of β-D-glucose is 64.3.

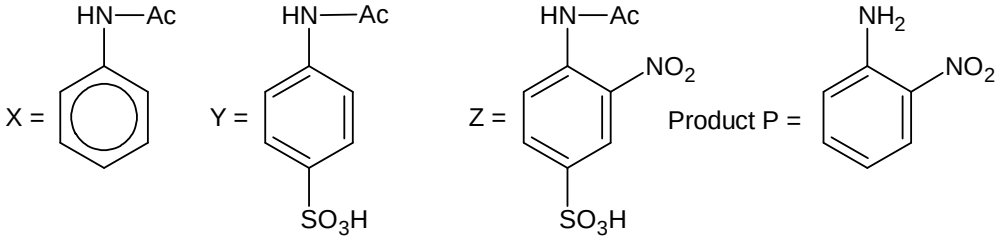
9. Leaving group is at β position to anion stabilizing group (i.e.  group.)

11.



12. All others are used for this purpose and they form cyclic acetal or cyclic thioacetals at  group.

17. Bauxite is ($Al_2O_3.2H_2O$)
Magnetite is (Fe_3O_4)
Bornite is (Cu_3FeS_3)
Cerussite is ($PbCO_3$)

21. 

22. $[NiCl_4]^{2-} \Leftarrow dsp^2$, square planer and diamagnetic
$[Cu(NH_3)_4]^{2+} \Leftarrow dsp^2$, square planer and paramagnetic
$[Ni(CO)_4] \Leftarrow$ is sp^3, tetrahedral but diamagnetic

25. $\kappa_{CaF_2} = \kappa_{CaF_2}$ (solution) - κ_{H_2O}
$= 4.05 \times 10^{-3}$ S/m
$\text{Conc} = \frac{\kappa}{\Lambda_m} = 2.025 \times 10^{-4}$ mol / dm^3
$K_{sp} = 4C^3 = 3.32 \times 10^{-11} M^3$

27. Heat supplied at constant pressure in the range of temperature, ΔT is ΔH and change in internal energy is ΔU hence $\frac{\Delta U}{\Delta H} = \frac{1}{\gamma} = \frac{1}{1.4} = 0.71(\text{approx})$

28. (A) $2KClO_3 + I_2 \longrightarrow 2KIO_3 + Cl_2$
(B) $5NaBr + NaBrO_3 + 6HCl \longrightarrow 6NaCl + 3Br_2 + 3H_2O$
(C) $CaOCl_2 + 2NaI + 2HCl \longrightarrow I_2 + CaCl_2 + H_2O + NaCl$
(D) $NaCl + H_2SO_4 \longrightarrow NaHSO_4 + HCl$

29. A and C possesses plane of symmetry. While D possesses centre of symmetry.

Mathematics

PART – III

SECTION – A

1. ${}^nC_3 = {}^nC_2 \Rightarrow n = 5$

2. $\frac{|PA| + |PB|}{2} > [|PA| \cdot |PB|]^{1/2}$

   $|PA| + |PB| > 2|PT| = 2\sqrt{3}$

   Maximum length occurs when PAB passes through centre

   i.e. $|PA| + |PB| = 4$ (Maximum)

   So, range is $\left(2\sqrt{3},\ 4\right]$

3. Length of tangent is = $\sqrt{|z_1 - i|^2 - 4}$

   $= \sqrt{|3 - 2i|^2 - 4} = \sqrt{9 + 4 - 4} = 3$

4. Required number of sentences = $\left({}^6C_0 + {}^6C_1 + {}^6C_2\right) \times \frac{7!}{2!}$

5. The given line can be written as $y = x$

   Hence reflection is $3 + 2i$

6. $\frac{1}{a_{n+1}} = \frac{1}{a_n(a_n + 1)} = \frac{1}{a_n} - \frac{1}{a_n + 1}$

   $\Rightarrow S = \frac{1}{a_1} - \frac{1}{a_2} + ... + \frac{1}{a_{100}} - \frac{1}{a_{101}}$

   $= \frac{1}{a_1} - \frac{1}{a_{101}} = 2 - \frac{1}{a_{101}}$

   Since, $a_{101} > 1 \Rightarrow [S] = 1$

7. Height of the cone = $\frac{5}{3}$

   Radius of cone = $\sqrt{9^2 - \left(\frac{5}{3}\right)^2} = \frac{8}{3}\sqrt{11}$

   Volume = $\pi\left(\frac{8}{3}\sqrt{11}\right)^2\left(\frac{5}{3}\right) = \frac{3520}{27}\pi$

8. $4! = 24$

   $5! = 24 \times 5$

   $6! = 24 \times 5 \times 6$

   $1! + 2! + 3! = 9$

   So, $1! + 2! + 3! + ... 100! = 9 + 24p$

   $\therefore \lambda = 9$

9. $|z| = O \times O - I \times I = -I^2$

   $\det(|z|) = |-I^2| = (-1)$

10. $f(x)+f\left(\frac{1}{1-x}\right)=\frac{2}{x}-\frac{2}{1-x}$ ..... (1)

Now replace x by $\frac{1}{1-x}$ we get $f\left(\frac{1}{1-x}\right)+f\left(1-\frac{1}{x}\right)=2(1-x)-2\left(1-\frac{1}{x}\right)=-2x+\frac{2}{x}$ ..... (2)

Replace x by $1-\frac{1}{x}$ we get $f\left(1-\frac{1}{x}\right)+f(x)=\frac{2x}{x-1}-2x$ ..... (3)

From equation (1) – (2), we get

Now, $f(x)-f\left(1-\frac{1}{x}\right)=2x-\frac{2}{1-x}$ ..... (4)

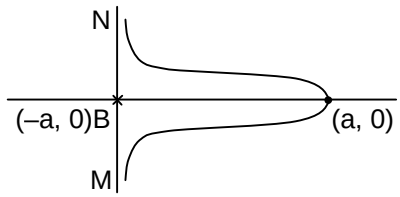
From equation (3) + (4), we get

Again, $2f(x)=\frac{2x}{x-1}-\frac{2}{1-x}$

$f(x)=\frac{x+1}{x-1}$ which is a one–one function

11. x = –a is the only asymptote to the given curve

$A=2\int_{-a}^{a} y\,dx=3\pi a^2$



12. $M=e^y$, $N=x\cdot e^y+2y$

Now, $\frac{dM}{dy}$ (keeping x constant) $=e^y$

$=\frac{dN}{dx}$ (keeping y constant)

Hence, the given equation is exact

Now, $f=\int e^y dx+g(y)=x\cdot e^y+g(y)$

Differentiating w.r.t y we get, $x\cdot e^y+\frac{dg}{dy}=N=x\cdot e^y+2y$

i.e. $\frac{dg}{dy}=2y$

Integrating $g(y)=y^2+c_1$

$\Rightarrow f=x\cdot e^y+y^2+c_1$

So, desired solution is $x\cdot e^y+y^2=0$

13. $\bar{b}\times(\bar{v}\times\bar{b})=\bar{b}\times\bar{c}$

$4\bar{v}-(\bar{b}\cdot\bar{v})\bar{b}=\bar{b}\times\bar{c}$

$|\bar{v}\times\bar{b}|=|\bar{c}|$

$\Rightarrow |\bar{v}||\bar{b}|\sin\theta=|\bar{c}|$

$2|\bar{v}|\sin\theta=\sqrt{3}$

$\sin\theta=\frac{\sqrt{3}}{2}$

$\cos\theta = \frac{1}{2}$

$\Rightarrow 4\bar{v} - \left(2\times1\times\frac{1}{2}\right)\bar{b} = \bar{b}\times\bar{c}$

$\bar{v} = \frac{1}{4}(\bar{b} + \bar{b}\times\bar{c})$

14. Area under curve and axis = $\int_0^1 (1-x^2)dx = \frac{2}{3}$ at point $(\alpha, 1-\alpha^2)$ tangent is

$[y - (1-\alpha^2)] = -2\alpha(x-\alpha)$

$\Rightarrow 2\alpha x + y = \alpha^2 + 1 \Rightarrow P \equiv \left(\frac{\alpha^2+1}{2\alpha}, 0\right)$, $Q \equiv (0, \alpha^2+1)$

Hence, $Q \equiv \left(2, 1+4x_1 - x_1^2\right)$

$\therefore A = \left[\left(1+2x_1 - x_1^2\right) + \left(1+4x_1 - x_1^2\right)\right] = 1 - 3x_1 - x_1^2$

$\frac{dA}{dx_1} = -3 - 2x_1 \Rightarrow x_1 = -\frac{3}{2}$

15. $\left(-\frac{\alpha}{4}, 0\right)$ lies on directrix and centre be $(x_1, 0)$. Equation of one of the tangent is $x + y + \frac{\alpha}{4} = 0$ and chord is $x - \frac{\alpha}{4} = 0$, centre is $(x_1, 0)$

$\frac{\left|x_1 + \frac{\alpha}{4}\right|}{\sqrt{2}} = \frac{\left|x_1 - \frac{\alpha}{4}\right|}{1} \Rightarrow x_1^2 + \frac{\alpha^2}{16} + \frac{\alpha x}{2} = 2x_1^2 + \frac{\alpha^2}{8} - \alpha x$

$x_1^2 - \frac{3\alpha x}{2} + \frac{\alpha^2}{16} = 0 \Rightarrow x_1 + x_2 = \frac{3\alpha}{2} \Rightarrow$ Midpoint $\equiv \left(\frac{3\alpha}{4}, 0\right)$

16. $2x + y - 1$ is the axis and $(1, -1)$ is the point of intersection with the tangent at the vertex, length of latus rectum is $\sqrt{5}$

So, coordinates can be (h, k)

$h = 1 \pm \left(\frac{\sqrt{5}}{4}\right)\left(-\frac{1}{\sqrt{5}}\right) = 1 \mp \frac{1}{4} = \frac{3}{4}, \frac{5}{4}$

$k = -1 \pm \left(\frac{\sqrt{5}}{4}\right)\left(\frac{2}{\sqrt{5}}\right) = -1 \pm \frac{1}{2} = -\frac{1}{2}, -\frac{3}{2}$

$\left(\frac{3}{4}, -\frac{1}{2}\right)$ and $\left(\frac{5}{4}, -\frac{3}{2}\right)$ are required points

17. For the common tangent $m^2 \sin^2\theta + \cos^2\theta = m^2 + 1$

$m^2(\sin^2\theta - 1) = 1 - \cos^2\theta$

$\Rightarrow m^2 = \frac{1-\cos^2\theta}{\sin^2\theta - 1} = -\frac{\cos^2\theta}{\sin^2\theta} = -\cot^2\theta$

Clearly no such m is possible (when $\left(\frac{\pi}{4} < \theta < \frac{\pi}{2}\right)$)

18. Let midpoint is (h, k) then chord is $y = -\frac{h}{k}x + 2h$ and if it is a tangent then $x^2 = 4\left(-\frac{h}{k}x + 2h\right)$

D = 0

$\Rightarrow k^2 = -\frac{h}{2}$ (Parabola)

19. $\left|\alpha_1\beta_1 + \alpha_2\beta_2\right| \le \sqrt{\alpha_1^2 + \alpha_1^2}\sqrt{\beta_1^2 + \beta_2^2}$

$\Rightarrow \left|\alpha_1\beta_1 + \alpha_2\beta_2\right| \le 1$

20. $g(n) = \int_0^{n^2+n+1} e^{\{x/2\}}\left\{\frac{x}{2}\right\}d\{x\}$

$= (n^2 + n + 1)\int_0^1 e^{\{x/2\}}\left\{\frac{x}{2}\right\}dx$

$= (n^2 + n + 1)\int_0^1 e^{x/2}\left(\frac{x}{2}\right)dx$

$= n^2 + n + 1[4 - 2e^{1/2}]$

So, minimum value is $12 - 6\sqrt{e}$

21. f(0) < f(1) now if suppose

$\Rightarrow$ f(n – 1) < f(n)

$\Rightarrow \sqrt{f(n-1)+6} < \sqrt{f(n)+6}$

$\Rightarrow$ f(n) < f(n + 1)

Also, f(0) < 3, let f(n) < 3

$\Rightarrow \sqrt{f(n)+6} < \sqrt{3+6}$

$\Rightarrow$ f(n + 1) < 3

So, f(n) is bounded above 3

22. Any plane containing the line is $p_1 + \lambda p_2 = 0$. It's distance (d) from origin $= \frac{|1+\lambda|}{\sqrt{\sum(a+\lambda a')^2}}$.

Let $d^2 = \phi = \frac{\lambda^2 + 2\lambda + 1}{A\lambda^2 + B\lambda + C}$ where $A = \sum a'^2$, $B = 2\sum aa'$ and $C = \sum a^2$.

$\Rightarrow (A\phi - 1)\lambda^2 + (B\phi - 2)\lambda + (C\phi - 1) = 0$. As $\lambda \in R$, $D \ge 0$

$\Rightarrow 0 \le \phi \le \frac{4(A + C - B)}{4AC - B^2} = \frac{p}{q}$. So, $d_{max} = \sqrt{\frac{p}{q}}$.

23. We have $f(x + a) \ge \frac{1}{2}$ and so $f(x) \ge \frac{1}{2}$ $\forall$ x $\in$ R.

Lets set, $g(x) = f(x) - \frac{1}{2}$ we have $g(x) \ge 0$ $\forall$ x $\in$ R

$\Rightarrow g(x+a) = \sqrt{\frac{1}{4} - (g(x))^2}$

$\Rightarrow g^2(x+a) = \frac{1}{4} - g^2(x)$ = $g(x+2a) = \frac{1}{4} - g^2(x+a)$

$\Rightarrow g^2(x+2a) = g^2(x)$
$\Rightarrow g(x + 2a) = g(x)$
$\Rightarrow f(x + 2a) = f(x)$

24. $(\lambda I_n)\ Adj(\lambda I_n) = |\lambda I_n|\ I$
$\Rightarrow Adj(\lambda I_n) = \lambda^{n-1} I_n$
$\Rightarrow |Adj(\lambda I_n)| = \lambda^{n(n-1)}$

25. $B = A_1 + 3(A_2)^3 + ... + 13(A_7)^{13}$
$B^T = -[A_1 + 3(A_2)^3 + ... + 13(A_7)^{13}] = -B$

26. $$f = (a+p)^{m-1}\left[1+\frac{a+q}{a+p}+\left(\frac{a+q}{a+p}\right)^2+.....+\left(\frac{a+q}{a+p}\right)^{m-1}\right]$$
Now coefficient of a^t is $^mC_t \dfrac{p^{m-t}-q^{m-t}}{p-q}$

27. Point of intersection of L and x = –2 is the required point

28. $$\int_0^{\ln 2} e^{-2x} f''(x)\,dx$$
$$= \left[e^{-2x} f'(x)\right]_0^{\ln 2} + 2\int_0^{\ln 2} e^{-2x} f'(x)\,dx$$
$$= \left[\frac{1}{4}f'(\ln 2) - 3\right] + 2\left[e^{-2x}f(x) + 2\int e^{-2x} f(x)\,dx\right]_0^{\ln 2} = 13$$

30. $|a - b| < c$
$|b - c| < a$
$|c - a| < b$
$$\left.\begin{aligned}(a-b)^2 &< c^2\\ (b-c)^2 &< a^2\\ (c-a)^2 &< b^2\end{aligned}\right\} \Rightarrow (a-b)^2+(b-c)^2+(c-a)^2 < a^2+b^2+c^2$$
$\Rightarrow a^2 + b^2 + c^2 < 2ab + 2bc + 2ca$
$\Rightarrow a^2 + b^2 + c^2 + 2ab + 2bc + 2ca < 4(ab + bc + ca)$
$\Rightarrow 4(ab + bc + ca) - (a + b + c)^2 > 0$
Also, $2^a + 2^{-a} \geq 2$
Hence, for all values of a, b, c the above z is defined.